

Matroids

Def'n: E be a finite set
 $\mathcal{I} \subseteq 2^E$, $\mathcal{I} \neq \emptyset$.

$\mathcal{M} = (E, \mathcal{I})$ is called a matroid if:

(i) $I \subseteq J$, $J \in \mathcal{I} \Rightarrow I \in \mathcal{I}$

"Downward Closed"

(ii) $I, J \in \mathcal{I}$, $|I| < |J|$

$\Rightarrow \exists z \in J \setminus I$ with $I + z \in \mathcal{I}$.

Terminology: $\mathcal{I}$ is the set of "independent sets"

Basis, Rank, Circuit, Loop

Given $M = (E, \mathcal{I})$, then for $U \subseteq E$

a subset $B \subseteq U$ is called a base if
it is an inclusionwise maximal ~~ind~~ subset of
 U that is independent, i.e., $B \in \mathcal{I}$.

Lemma: Any two bases of U have the same cardinality.

This allows us to define the rank function.

For $U \subseteq E$, $r_M(U) = \text{size of a base of } U$

A subset $U \subseteq E$, spans E if it contains a base.

A circuit is an inclusionwise minimal dependant set.

A loop is a element set that is a circuit.

An Equivalent Definition

Thm: For $E, \mathcal{I}$ satisfying downward
closed property, $(E, \mathcal{I})$ is a

matroid iff:

(ii') $I, J \in \mathcal{I}$ s.t. $|I \setminus J| = 1$,
 $|J \setminus I| = 2$, then $I + z \in \mathcal{I}$ for some
 $z \in J \setminus I$.

pf: (ii) $\rightarrow$ (ii') clear.

(ii') $\rightarrow$ (ii) Exercise.

Examples of Matroids — 1/2

- * Linear Matroid: A matroid $M = (E, \mathcal{I})$ is a linear matroid if $\exists$ vectors $v_1, \dots, v_n$, $n = |E|$, and $v_i \in \mathbb{R}^q$ or some other vector space, such that $I \in \mathcal{I} \iff \{v_i : i \in I\}$ are linearly ind. Ex: show that for any collection of $v_1, \dots, v_n \in \mathbb{R}^q$, and $\mathcal{I} = \text{lin ind subsets}$, this is a matroid.
- * Graphic Matroid: Let $G = (V, E)$, let $\mathcal{I}$ be the subsets of E that form forest. Ex: show that this is a matroid.
- * Binary Matroid is a linear matroid over $\mathbb{F} = \text{GF}(2)$.

Examples of Matroids — 2/2

* Uniform matroid: $M = (E, \mathcal{I})$, $\mathcal{I}$ = all subsets of
cardinality $\leq k$.

* Partition matroid: $E = E_1 \cup E_2 \cup \dots \cup E_\ell$, disjoint
unions.
and $k_1, \dots, k_\ell \in \mathbb{Z}_+$.

$$\mathcal{I} = \{I \subseteq E : |I \cap E_j| \leq k_j, \forall j\}$$

* Transversal Matroid: $E_1, \dots, E_\ell \subseteq E$, not nec. disjoint,
 $\mathcal{I} = \{I \subseteq E : I = (x_1, \dots, x_k), \text{ then } x_j \in E_j, \text{ with}$
 $i_1, i_2, \dots, i_k \text{ disjoint.}$

* Matching Matroid

$G = (V, E)$, $M = (V, \mathcal{I})$, $\mathcal{I} = \{I \subseteq V : I \text{ covered by}$
some matching in $G\}$

